

Interstitial Lung Disease in a Dental Technician Diagnosed Through an Integrated Multidisciplinary Approach

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SUMMARY

Evidence from the literature indicates that workers in the dental care sector are routinely exposed to hazardous airborne agents, particularly mineral and metal dusts. This occupational risk is further amplified in settings with inadequate ventilation and insufficient use of personal protective equipment. We discuss the case of a dental technician with a significant history of occupational exposure who presented to our clinic with exertional dyspnea. High-resolution computed tomography revealed fibrosing nonspecific interstitial pneumonia associated with emphysema, and spirometry showed a severe reduction in carbon monoxide diffusion capacity. After a multidisciplinary evaluation, we performed bronchoscopy with bronchoalveolar lavage (BAL), which revealed an abundance of macrophages. Scanning electron microscopy with energy-dispersive spectrometry of the BAL detected silicon oxide, calcium, iron, and traces of metals typical of dental alloys (titanium, aluminum), as well as rare earth elements and toxic pollutants. These findings supported the diagnosis of interstitial lung disease due to long-term occupational exposure in dental laboratories, enabling a minimally invasive diagnostic approach. This strategy obviated the need for transbronchial biopsy, thereby avoiding a procedure associated with a high risk of pneumothorax in this patient.

1. INTRODUCTION

Dental care workers are exposed to a variety of physical, chemical, and biological hazards in their work environment. Dental technicians, in particular, are exposed to numerous potentially toxic

substances and airborne residues, such as silica, alloys, and acrylic plastics, during their work. As a result, they are at risk of developing occupational lung diseases, including pneumoconiosis, with severity often depending on the duration of exposure [1].

Research by Ali et al. in Saudi Arabia found limited awareness among dental technicians and dental technology students of the risks associated with exposure to dental material dust [2]. Most self-employed dental technicians work more than eight hours a day without adequate ventilation, increasing their risk of occupational lung conditions [3]. When ventilation is inadequate, dust particles, particularly those generated during grinding, have been shown to significantly exceed safety standards [4].

Pneumoconiosis is a subset of occupational ILDs characterized by the accumulation of inhaled inorganic particles in the lungs. It is one of the most prevalent occupational diseases globally, particularly in low- and middle-income countries [5, 6]. Based on the type of mineral dust responsible for the condition, pneumoconiosis can be classified into several categories: silicosis, the most common form, triggered by dust primarily containing free crystalline silica; silicate pneumoconiosis, caused by dusts containing silicates, including asbestos, talc, cement, and mica; pneumoconiosis resulting from coal dust and carbon-based dusts, such as coal worker's pneumoconiosis, graphite pneumoconiosis, and carbon black pneumoconiosis; and metal pneumoconiosis, caused by metal dusts, including aluminum pneumoconiosis, welder's pneumoconiosis, and caster's pneumoconiosis [7].

China accounts for more than half of the global burden of pneumoconiosis, representing 68.7% of incident cases, 44.3% of deaths, and 66.2% of disability-adjusted life years (DALYs). In 2019, 15,898 new cases were reported, and more than 20 million workers were exposed to occupational risk factors [8]. Recent reports from North-Eastern Italy have described an unexpectedly high incidence of silicosis among workers processing quartz conglomerates, a new artificial stone with high silica content used for kitchen and bathroom countertops, highlighting the occurrence of occupational pneumoconiosis in newer productive settings outside traditional mining and quarrying industries [9]. Currently, no curative treatment is available for pneumoconiosis, and most therapeutic approaches are aimed at slowing disease progression and alleviating symptoms [8].

Dental technicians' pneumoconiosis (DTP) is recognized as a distinct condition, often linked

to exposure to cobalt-chromium-molybdenum (Co-Cr-Mo) alloys [1]. A 2014 Turkish study of 893 dental technicians admitted to the hospital in the past 5 years found that 10.1% had pathological chest X-rays. The main HRCT abnormalities were hilar and mediastinal lymphadenopathy, micronodules, reticulonodular infiltration, increased linear density, and interlobular septal thickening. In this study, the highest rate of DTP was observed among individuals working in the leveling and polishing department. Among the subgroup with a history of sandblasting, pneumoconiosis was found in 46.1% of cases [1]. In another article by Kahraman et al., 46% of participants had radiological signs compatible with the condition, with the most common finding being round consolidations. In terms of pulmonary function, forced expiratory rate and diffusing capacity of the lungs for carbon monoxide (DLco) were significantly lower in those who had worked for 16 years or more [10].

In 2022, Di Lorenzo et al. described a case of pneumoconiosis in a dental technician with 15 years of experience, diagnosed by scanning electron microscopy-energy dispersive spectrometry (SEM-EDS) of a surgical biopsy. The procedures he performed in his job, sometimes in non-closed-circuit systems and often without protective equipment, generated aluminum, crystalline silica, chromium, and titanium dust. HRCT of the chest showed bilateral micro nodulation, with a greater concentration in the upper lobes. Histological examination revealed aggregates of pigment-laden macrophages forming perivascular macules or arranged in a radial pattern around a core of sclerohyalinosis. SEM-EDS identified several mineral particles, typically characterized by the presence of crystalline silica and metal (aluminum, iron, chromium, cobalt, titanium) aggregates. The environmental concentrations of total dust and its respirable fraction were below the respective Threshold Limit Value – Time-Weighted Average (TLV-TWA) (according to the American Conference of Governmental and Industrial Hygienists – ACGIH), but still not negligible [11].

Among the few cases reported in the literature on mineralogical analyses of BAL, De Vuyst et al., in 1986, identified, by means of SEM/EDS, particles of

chromium, cobalt, and molybdenum—constituents of Vitallium®, a chromium–cobalt–molybdenum alloy—in the BAL of two patients with biopsy-proven pulmonary fibrosis. The authors also demonstrated the chemical stability of this alloy in BAL fluid and lung tissue several years after cessation of exposure. Indeed, in one patient, mineralogical analysis of BAL was performed both during occupational exposure and one year after cessation of work activity, showing similar findings. In the second patient, BAL analysis demonstrated the presence of Vitallium® particles two years after occupational exposure had ended. In the BAL of one patient, bare asbestos fibers as well as typical asbestos bodies were also detected, whereas in the other patient iron, iron–chromium, and silica particles were identified [12]. More recently, Tiraboschi et al. reported the case of a ceramic dental technician with a two-year occupational history who developed pneumoconiosis. SEM/EDS analysis of the patient's BAL revealed numerous inorganic particles containing zirconium and aluminum, as well as tungsten [13].

2. CASE PRESENTATION

We discuss the case of a 69-year-old male patient born in 1956, a non-smoker who was exposed to passive smoke for 5 years, from 1973 to 1978. He was a dental technician who worked in the field, exposed to ceramic and plaster dust, monomers, and polymers, particularly from ages 18 to 23. Subsequently, he was provided with personal protective equipment and extraction hoods in his workplace. However, regarding plaster, the patient specified that the material undergoes an initial processing phase during which the operator does not use personal protective equipment. His grandfather had an unspecified lung disease. As comorbidities, the patient reported only type 2 diabetes and arterial hypertension. He denied early graying. As home therapy, the patient was taking metformin and tamsulosin.

In 2020, the patient began experiencing shortness of breath during intense exertion (modified Medical Research Council – mMRC – Dyspnea Scale 0). On the general practitioner's advice, he underwent cardiopulmonary exercise testing (CPET), which was negative, and pulmonary function tests (PFTs),

which showed normal lung volumes and a moderate reduction in the DLco to 45%.

The patient discontinued follow-up until 2024, when persistent respiratory difficulty prompted further evaluation. On a pulmonologist's recommendation, he underwent high-resolution computed tomography (HRCT) of the chest, which revealed a pattern of fibrosing nonspecific interstitial pneumonia (fNSIP) associated with emphysema (Figure 1). He also had new PFTs, showing a DLco of 34% and a mild restrictive defect, with a total lung capacity (TLC) of 78%. An echocardiogram showed normal pump function, with an estimated systolic pulmonary artery pressure of 28 mmHg, below the commonly proposed echocardiographic thresholds for pulmonary hypertension.

At our first visit, the patient was eupneic at rest on ambient air, with peripheral oxygen saturation of 98%. Upon auscultation of the chest, vesicular murmurs were reduced throughout the lung fields, with rare crackles at the bases. Digital clubbing was noted.

We recommended a blood sample for antibody panel testing, which was entirely normal except for a weakly positive rheumatoid factor (53.9 kU/L) in the absence of clinical features suggestive of connective tissue disease. Therefore, this finding was considered to be of no pathological significance. Moreover, alpha-1 antitrypsin levels were tested and found to be within the normal range. The case was then discussed at the multidisciplinary meeting dedicated to interstitial and rare lung diseases. The fNSIP radiological pattern was confirmed, with ground-glass areas, particularly in the lower lobes, associated with paraseptal emphysema. Radiological findings typically associated with fibrotic hypersensitivity pneumonitis (fHP), particularly those reflecting a bronchiolocentric interstitial pneumonia (BIP) pattern, such as mosaic attenuation, lobular air trapping, and the three-density sign, were not observed. Other differential diagnoses were carefully considered. No other relevant elements emerged from the history, apart from the well-documented long-standing occupational exposure commonly encountered in dental laboratories. Moreover, although not classically considered a predominant feature of these conditions, emphysematous changes have

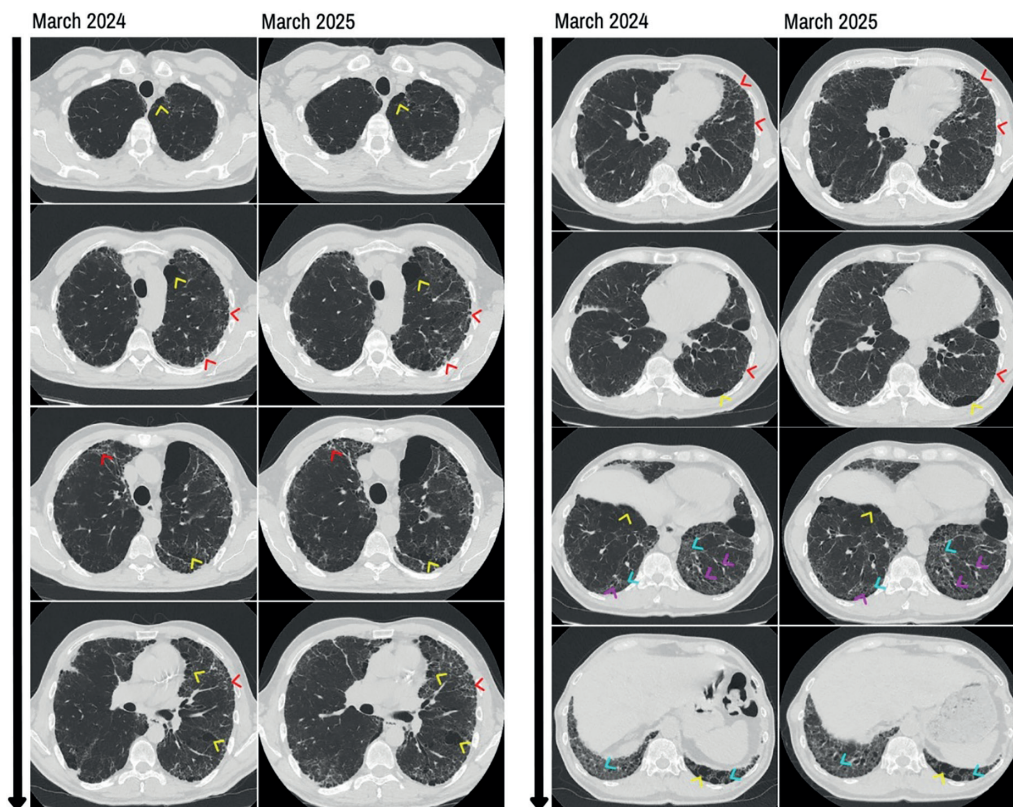


Figure 1. Chest HRCTs obtained one year apart. Findings include emphysematous changes with enlargement of the air spaces (yellow arrowheads), subpleural interstitial thickening (red arrowheads), areas of mild ground-glass opacity predominantly involving the lower lobes (light blue arrowheads), and traction bronchiectasis (purple arrowheads). A mild radiological progression is observed from March 2024 to March 2025.

been described in several forms of pneumoconiosis associated with inorganic dust exposure and were therefore considered compatible with the patient's occupational history. Considering the overall clinical, radiological, and occupational picture, the disease was considered compatible with occupational pneumopathy rather than an idiopathic process.

From the consultation with the occupational physicians, additional details emerged regarding the patient's work history. He worked as a dental technician from 1973 to 1990. During the early years as an apprentice, he mixed plaster with water and poured wax to create models for removable prostheses. Later, he worked in the production of acrylic resin, derived from a mixture of liquid monomer

and powdered polymer. The acrylic prosthesis was then deflashed (removed from the plaster mold) and baked at 90°C. This was followed by milling, first polishing with pumice stone and final polishing with brushes. These procedures involved inhalation of fumes and dust from the materials being worked. Since 1990, he has been involved in orthodontics, with work activities similar to those above but characterized by reduced exposure to dust and fumes (including those from welding the tempered steel wires of prostheses with a butane torch, a very sporadic activity) due to the constant use of direct suction systems. From the technical data sheets of the products used, it was found that the materials contained gypsum (calcium sulfate dihydrate), iron

oxides, titanium oxides, and methacrylate (in orthodontic resins), as well as nickel, chromium, and manganese in the stainless-steel wires and screws used in orthodontics, and therefore present in welding fumes, although the welding activity was reported as very sporadic. The use of pumice stone also exposed him to silica and aluminum. Thus, the occupational history documented multiple exposures and strengthened suspicion of a possible causal role of work in the onset of the radiological alterations previously described.

For further diagnostic purposes, the patient underwent a bronchoscopy with BAL, whose cytologic analysis showed a predominance of macrophages (80%), many of which were pigmented, with a significant proportion demonstrating activation (Figure 2). SEM-EDS on the BAL revealed the presence of silicon oxide, calcium, iron, and traces of metals typical of dental alloys and pumice stone (titanium, aluminum), as well as rare earth elements such as terbium, in addition to toxic pollutants like antimony (Figure 2). Transbronchial biopsies have been avoided due to the risks associated with the procedure, first pneumothorax, given the condition of lung parenchyma.

Combined radiological and functional findings, along with BAL analysis using SEM-EDS that identified minerals and metals consistent with those present in the products used, allowed us to confirm the occupational etiology of the pneumopathy affecting the patient. Moreover, BAL analysis did not demonstrate lymphocytosis, a finding that further argued against fHP. Based on the mineralogical findings, this condition can be classified as an occupational ILD compatible with mixed-dust exposure, presenting with a fNSIP pattern. The patient was advised to cease all work activities involving direct exposure to the dust and fumes mentioned above and has been placed under clinical, radiological, and functional follow-up.

The patient discontinued work-related activities; however, in March 2025, radiologic (shown in Fig. 1) and functional disease progression was observed, with a 7% decline in forced vital capacity compared with the previous year. In light of the observed disease course, antifibrotic therapy with

nintedanib was initiated to slow disease progression. By July 2025, the patient developed reduced exercise tolerance; the 6-minute walk test demonstrated oxyhemoglobin desaturation occurring at the second minute of exercise, prompting the initiation of exertional oxygen therapy.

3. CONCLUSION

To the best of our knowledge, this is one of the few documented cases of ILD with an occupational etiology in a dental technician, confirmed by SEM-EDS analysis of BAL. We find the use of this technique and its diagnostic yield on endobronchial fluid particularly interesting, as it allowed us to avoid a biopsy in a patient for whom such a procedure would have been highly risky. Consequently, we achieved a definitive diagnosis with high confidence using an invasive procedure that can still be safely performed even in high-risk patients. The combination of chronic occupational exposure to multiple inorganic dusts and metals and BAL SEM-EDS findings strongly supported the occupational etiology of the ILD. In this context, we considered the case to be more appropriately classified as an occupational ILD compatible with mixed-dust exposure. The coexistence of fibrotic interstitial features and documented exposure to heterogeneous inorganic particles may also place this condition within the spectrum of mixed-dust pneumoconiosis.

The fibrogenic potential of the substances identified by BAL SEM-EDS analysis is likely heterogeneous. In our opinion, the occupational relevance of this case is primarily related to chronic inhalational exposure to inorganic dusts and metals with recognized pneumotoxic and fibrogenic potential, particularly silica-containing materials and iron particles, followed by aluminum, and, although less commonly reported, titanium particles [11], all of which were consistent with the patient's occupational history. In contrast, elements such as antimony and rare earth elements were interpreted as ancillary findings detected by the high sensitivity of SEM-EDS analysis rather than as primary drivers of the fibrotic lung disease.

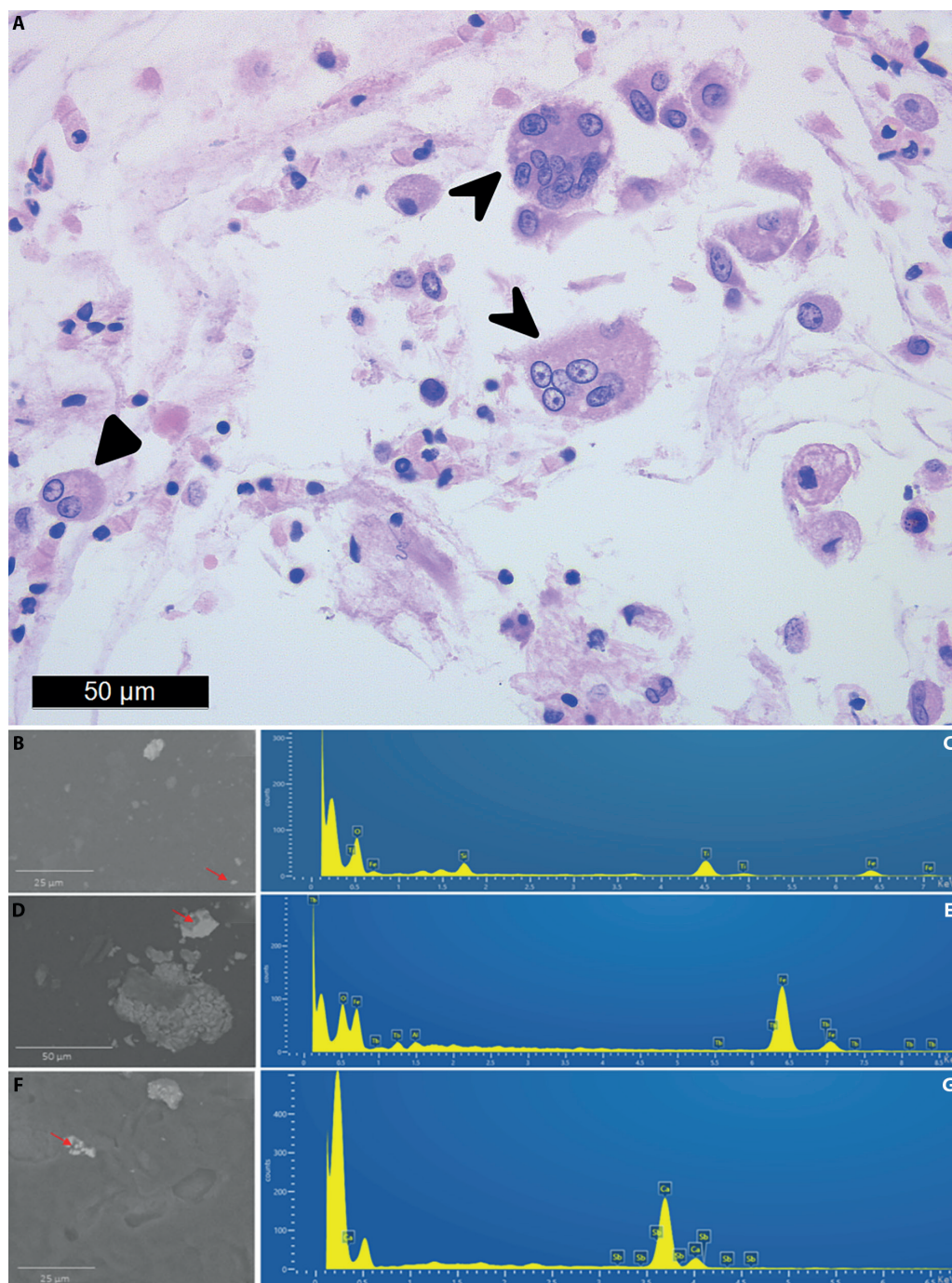


Figure 2. Bronchoalveolar lavage cytoblock H&E slides (A) demonstrated an abundant amount of macrophages. A significant proportion of these (between 15-20%) demonstrated activation (notched arrowhead); another 5% were multinucleated macrophages (full arrowhead). SEM-EDS on BAL paraffin sections (B-G): B) A particle (arrow) contains silicon, iron and titanium (C); D) in a particle (arrow) aluminum, iron, terbium are detected (E); F) A particle (arrow) contains calcium and antimony (G).

This case report aims to draw attention to this category of workers, highlighting how occupational exposures in these environments can harm the lungs. Moreover, it seeks to raise awareness of the importance of using personal protective equipment and implementing adequate suction and ventilation systems to improve workplace health and safety. Last but not least, our patient was a freelancer and did not undergo regular health surveillance examinations provided by an occupational physician; thus, the lack of periodic monitoring has reasonably delayed the diagnosis. Pneumoconiosis is a completely preventable but irreversible disease; for this reason, it is necessary to strengthen awareness among workers about occupational risk management tools and the importance of undergoing regular occupational health surveillance.

Additionally, we believe that bronchoscopy with BAL and SEM-EDS analysis may be a valuable diagnostic tool in selected cases of suspected occupational ILD. A potential surveillance strategy for at-risk individuals, such as those employed in the dental care sector, could initially rely on non-invasive approaches, including PFTs, potentially supplemented by lung ultrasound when an experienced pulmonologist with expertise in ILD is available within the screening setting. Subjects with findings suggestive of interstitial involvement could then undergo chest HRCT for further evaluation. Finally, in cases where ILD is confirmed and additional etiological characterization is clinically indicated, bronchoscopy with BAL followed by SEM-EDS analysis could be integrated into the standard multidisciplinary diagnostic work-up to support correlation of lung pathology with occupational exposure.

INFORMED CONSENT STATEMENT: Informed consent was obtained from the subject involved in the study. Written informed consent has been obtained from the patient to publish this paper.

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AUTHOR CONTRIBUTION STATEMENT: GG was the first author and was responsible for drafting the manuscript, reviewing the patient's clinical data, and conducting the literature review; GM analyzed the bronchoalveolar lavage samples and contributed to drafting the manuscript. FC led these analyses, drawing on her extensive expertise in pathology; MD contributed by performing the endoscopic examination that led to the final diagnosis; RP reviewed and interpreted the radiological findings as an expert thoracic radiologist; SR, MDB, and CB contributed their domain expertise to the scanning electron microscopy and energy-dispersive X-ray spectroscopy analyses; PM contributed to the comprehensive occupational and medical history assessment as an expert in occupational medicine and provided valuable input in the complex clinical management of the case; PS provided expert pneumological input during manuscript preparation; EB coordinated the work and provided expert guidance in drafting and revising the manuscript. All authors have read and approved the final version of the manuscript.

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